

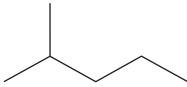
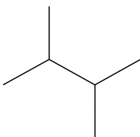
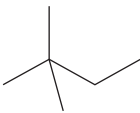
7. Petroleum ether (50–70) is a mixture of different alkanes extracted from crude oil which is commonly used as an organic solvent. The major components of petroleum ether (50–70) are the structural isomers of C_6H_{14} .

(a) (i) Give the meaning of the term 'structural isomer'. [1]

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(ii) Complete the table below showing important information about the isomers of C_6H_{14} . [3]

Name	Shortened structural formula	Skeletal formula	Boiling temperature / °C
hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$		69
2-methylpentane			62
3-methylpentane	$CH_3CH_2CH(CH_3)CH_2CH_3$		63
	$(CH_3)_2CHCH(CH_3)_2$		58
2,2-dimethylbutane	$CH_3C(CH_3)_2CH_2CH_3$		50



- (iii) State the relationship between the boiling temperature and the carbon chain length. Explain this relationship in terms of intermolecular forces. [2]

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- (b) Hexane can be used as a fuel in a combustion reaction.

- (i) Write an equation for the complete combustion of hexane in excess oxygen. [2]

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- (ii) The enthalpy change of combustion ($\Delta_c H^\theta$) for hexane is approximately $-4160 \text{ kJ mol}^{-1}$. Explain why the enthalpy change of combustion for the isomers of hexane should be similar. [2]

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- (iii) 2,2-dimethylbutane is the isomer of C_6H_{14} which ignites most readily. Suggest a reason for this. [1]

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- (iv) When hexane burns in a limited supply of oxygen it undergoes a different reaction known as incomplete combustion:



The bond enthalpy values for the bonds present in these molecules are given below:

Bond	Average bond enthalpy / kJ mol^{-1}
$\text{C} - \text{C}$	348
$\text{C} - \text{H}$	413
$\text{O} = \text{O}$	495
$\text{C} \equiv \text{O}$ (in CO)	1072
$\text{O} - \text{H}$	464

- I. Using a Hess cycle or otherwise, calculate the enthalpy change of this reaction.

[3]

enthalpy change = kJ mol^{-1}

- II. **Use the enthalpy values** from parts (b)(ii) and (b)(iv) I. to explain quantitatively why it is important to maintain an excess of oxygen while burning hexane as a fuel.

[2]

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III. State a health hazard associated with the incomplete combustion of hexane. [1]

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